COMP364 Quiz 5

Question 1. Describing the value-learning algorithm (5 points)

(a) State the update equation of the value-learning algorithm studied in class.

Solution.

$$V'(s_t) = V(s_t) + \alpha\big(\widetilde{V}(s_{t+1}) - V(s_t)\big)$$

Explanation.

We work on *game graph*: a directed graph containing a start node (usually a) and end node (usually z). Each node n is labelled with a numeric *estimated value* $V(n)$ and implicitly has a *true value* $V^*(n)$. We usually abbreviate "estimated value" to "value."

We imagine two players playing a game in which player one begins at $s_1 = a$, and chooses a successor state (i.e. a node) s_2 by following an edge from s_1. Player two then chooses a successor state s_3 by following an edge from s_2, and so on. The player who successfully chooses z is the winner. Winning a game is defined to have a value or reward of 1.0.

For node n, the true value $V^*(n)$ is the expected value of the reward for the current game for a player about to move *from* node n, assuming some particular strategy by each player. (In reinforcement learning, these strategies are called the *policy*.) The graph labels $V(n)$ are an estimate of $V^*(n)$. Our algorithm, which we simply call the *value-learning algorithm*, computes estimates of the true values for a policy in which each player uses optimal play.

Recall $V(n)$ is an estimated value for a player moving *from* n. We define $\widetilde{V}(n)$ an the estimated value for a player moving *to* n. The estimated values for the players must sum to 1, which is the total reward for winning the game, and exactly one player will win each game. Therefore, we have

$$\widetilde{V}(n) = 1 - V(n).$$

We have $V^*(z) = V(z) = 0$, because a player moving from z has already lost. Equivalently, $\widetilde{V}(z) = 1$, because a player moving to z wins the game.

An *episode* of the algorithm is a complete game, technically defined by a path in the game graph beginning at a and ending at z. Each episode has certain other pieces of data associated with it, such as whether each move was exploratory or greedy.

The value learning algorithm proceeds according to Sidebar 9.1 in *Thinking AI*. Below are a few additional details.

The algorithm's two parameters are the exploration probability ϵ and the step size (or learning rate) α. When exploring, the algorithm does not consider the greedy move as a possibility. In

particular, if there is only one choice other than the greedy move, then no random number is used to choose the exploratory move.

The update equation for the estimated values can be rewritten as a convex combination of $V(s_t)$ and $\widetilde{V}(s_{t+1})$:

$$\begin{aligned} V'(s_t) &= V(s_t) + \alpha\big(\widetilde{V}(s_{t+1}) - V(s_t)\big) \\ &= (1-\alpha)V(s_t) + \alpha\widetilde{V}(s_{t+1}). \end{aligned}$$

(b) What phrase would a reinforcement learning expert use to describe our value-learning algorithm?

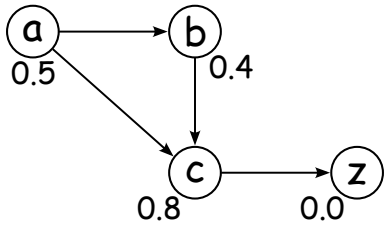
Solution.

It is a "tabular solution using temporal difference updates."

Question 2. Implementing the value-learning algorithm (10 points)

Show the outputs of two episodes of our value-learning algorithm for the two-player game shown below, assuming:

- the start state is a, and the final state is z;
- the current estimated values are as shown on the graph;
- $\epsilon = 0.3$ and $\alpha = 0.1$;
- the random numbers used by the algorithm, if needed, are $\rho_1 = 0.88, \rho_2 = 0.21, \rho_3 = 0.62, \rho_4 = 0.89, \rho_5 = 0.48$.



You must use the notation and method discussed in class.

Solution.

episode 1	a	$\rho_1 = 0.88 \xrightarrow{B}$	b	$\xrightarrow{B}$	c	$\xrightarrow{B}$	z
V	0.50		0.40		0.80		0.00
V'	0.51		0.38		0.82		

episode 2	a	$\rho_2 = 0.21 \xrightarrow{E}$	c	$\xrightarrow{B}$	z
V	0.51		0.82		0.00
V'			0.838		

Explanation.

Episode 1 begins in state a and uses the first random number, $\rho_1 = 0.86$, to decide between an exploratory or greedy move. Because $\rho_1 > \epsilon$, we use a greedy move, which means we move to state b—this has the lowest value for the opponent and therefore the highest value of for the current player.

The updated value $V'(a)$ is therefore

$$V'(a) = 0.5 + 0.1 \times (0.6 - 0.5).$$

Here we have used the fact that

$$\widetilde{V}(b) = 1 - 0.4 = 0.6.$$

There is only one choice for the next move, so the episode continues from node b to node c. The updated value estimate is

$$V'(b) = 0.4 + 0.1 \times (0.2 - 0.4) = 0.38,$$

where we have used the fact that

$$\widetilde{V}(c) = 1 - 0.8 = 0.2.$$

For the next move we again have only one choice, moving to z. The updated value for c is given by

$$V'(c) = 0.8 + 0.1 \times (1 - 0.8) = 0.82,$$

where we use the fact that

$$\widetilde{V}(z) = 1 - V(z) = 1 - 0 = 1.$$

Episode 2 proceeds as follows. It begins in state a and uses $\rho_2 = 0.21$, to decide between an exploratory or greedy move. Because $\rho_2 \leq \epsilon$, we use an exploratory move. No additional random number is consumed because there is only one option other than the greedy move. So we explore to c. The estimate $V(a)$ is not updated because the move was exploratory. Then we have only one choice which is to move to z. The updated value for c is given by

$$V'(c) = 0.82 + 0.1 \times (1 - 0.82) = 0.838,$$

where we again use the fact that $\widetilde{V}(z) = 1 - V(z) = 1 - 0 = 1$.

Total number of points: **15**